

ARGOS-Wound: A Multimodal Benchmark and Failure-Aware Fusion Framework for Open-Surgery Wound Perception

Controlled Ex-Vivo Surface Reconstruction, Semantic Understanding, and Uncertainty-Aware Evaluation for Robotic Surgical Guidance

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Abstract

Reliable local perception of the wound field is a prerequisite for any meaningful form of robotic guidance in open surgery. Unlike minimally invasive surgery, where perception is typically constrained to endoscopic views, open surgery presents a larger and less standardized workspace characterized by deformable tissue, severe hand and tool occlusions, wet and specular surfaces, blood, strong illumination changes, and heterogeneous anatomical layers. These factors make local 3D reconstruction and semantic understanding particularly challenging.

This document proposes **ARGOS-Wound**, a controlled ex-vivo benchmark for open-surgery wound perception. The benchmark will use porcine tissue to simulate open surgical wounds and will include synchronized multi-view RGB-D recordings, stereo imaging, active depth or ToF/LiDAR sensing, high-quality static anchor-state reference geometry, semantic annotations, wound-edge labels, tool and occlusion annotations, and baseline reconstruction methods. The goal is to create both a release-ready dataset and a rigorous evaluation protocol for studying passive stereo, active stereo, ToF/LiDAR sensing, multi-view reconstruction, and uncertainty-aware fusion in open surgical wound scenes.

A key design principle is that the first benchmark version will not attempt to provide dense dynamic ground truth for every frame. Instead, it will combine continuous synchronized sensing streams with high-quality static anchor geometry acquired at selected deformation states. Dense metric accuracy will be assessed primarily at anchor states, while dynamic clips will be evaluated through temporal consistency, visible-surface stability, occlusion recovery, and sensor failure-mode analysis.

Beyond dataset construction, ARGOS-Wound is intended to support **ARGOS-Fuse**, a semantic and uncertainty-aware multimodal fusion framework for visible wound-surface reconstruction. The central methodological goal is not only to reconstruct the wound surface, but also to estimate when and where the reconstruction should not be trusted. This failure-aware formulation is critical for future surgical guidance, where unsafe or unreliable perception should be explicitly detected rather than silently propagated to downstream robotic modules.

ARGOS-Wound is intended as a first perception-focused foundation for the broader ARGOS project, whose long-term goal is autonomous robotic guidance in open surgery. In its current form, this document should be read primarily as an internal de-risking proposal: its purpose is to determine whether open-wound perception can be measured, calibrated, annotated, and evaluated reliably enough to justify a larger benchmark and method-development effort targeting MICCAI 2027.

1 Purpose of This Internal Proposal

The purpose of ARGOS-Wound is to de-risk the perception layer of ARGOS before investing in more complex autonomy modules. The central question is not whether autonomous open surgery can already

be achieved, but whether local wound-field perception can be measured, calibrated, annotated, and evaluated reliably under controlled open-surgery-like conditions.

This proposal therefore defines a staged experimental plan to determine whether multi-view RGB-D sensing can provide sufficiently reliable semantic-geometric information for future robotic guidance tasks, such as wound-edge localization, tool–tissue distance estimation, deformation-aware reconstruction, and suturing assistance.

The outcome of Phase 1 should be a clear go/no-go decision for scaling the work toward a larger dataset, stronger baselines, an uncertainty-aware reconstruction method, and a top-tier submission. The main publication target, if Phase 1 succeeds, is **MICCAI 2027**.

2 Motivation

Autonomous robotic guidance in open surgery requires a perception system that can continuously understand the local surgical field. Before a robot can safely suture, retract, cut, drill, or assist the surgeon, it must estimate the geometry of the wound, identify visible tissue regions, localize wound edges, detect hands and tools, estimate tool–tissue distances, and decide when the visual scene is too uncertain or occluded for safe action.

Current surgical 3D reconstruction research is largely dominated by minimally invasive and endoscopic scenarios. These settings are important, but they differ substantially from open surgery. In open surgery, the wound is externally accessible, but the scene is also more variable and less constrained. The perception system must handle changing viewpoints, surgeon hands, metallic tools, retractors, suction tubes, gauze, blood, tissue deformation, and strong surgical illumination.

The first step toward ARGOS should therefore not be autonomous suturing itself. A more realistic and scientifically stronger first step is to build a controlled benchmark for open-surgery wound perception.

The central idea of ARGOS-Wound is to use an ex-vivo tissue setup where the experimental conditions can be controlled, sensors can be freely mounted, fiducials can be placed around the wound, and reference geometry can be acquired. This allows us to study the fundamental perception problem before moving to real operating-room recordings.

3 Core Research Question

The main research question is:

Can calibrated multi-view RGB-D sensing and semantic uncertainty-aware fusion provide reliable visible-surface reconstruction of an open surgical wound field while also detecting when and where the reconstruction is unreliable under tissue deformation, wetness, blood-like fluids, tool interaction, strong illumination, and severe occlusion?

This question decomposes into four subquestions:

1. Which sensing modality is most reliable for open-wound geometry: passive stereo, active stereo, ToF/LiDAR, or multi-view RGB?
2. Under which conditions do these sensors fail: wet tissue, blood, specular reflection, muscle/fat/fascia-like tissue, tools, or occlusion?
3. Can semantic information improve geometric reconstruction by masking tools, hands, background, and unreliable regions?
4. Can uncertainty-aware fusion outperform individual sensors and naive fusion not only in reconstruction accuracy, but also in failure detection, unsafe-region identification, and uncertainty calibration?

4 Staged Dataset Roadmap

ARGOS-Wound will be developed in stages. This staged design reduces the risk of overpromising dense dynamic ground truth before the calibration, reference-geometry, and annotation pipeline has been validated.

4.1 v0: Geometry-Anchored Stereo/RGB Benchmark

The v0 benchmark is the conservative and most defensible first release. It focuses on synchronized stereo/RGB acquisition, fiducial-based calibration, core semantic labels, and high-quality static anchor geometry.

The goal of v0 is not to solve the full dynamic deformation problem. Instead, it establishes whether open-wound perception can be measured reliably under controlled conditions.

The v0 dataset will include:

- synchronized stereo/RGB or RGB-D streams;
- global context video;
- rigid fiducials for calibration;
- static reference scans at selected anchor states;
- wound, tissue, hand, tool, fluid, gauze, background, and occlusion masks;
- visible-surface reconstruction evaluation at anchor states;
- temporal consistency and failure-mode analysis during dynamic clips.

The core geometry supervision in v0 will come from external static reference scans, not from ToF/LiDAR streams treated as absolute ground truth.

4.2 v1: Active Depth and ToF/LiDAR Extension

The v1 benchmark extends v0 by adding active depth and ToF/LiDAR sensing as measured modalities. These sensors will be evaluated against the anchor-state reference geometry and compared with passive stereo and multi-view RGB reconstruction.

The v1 dataset will include additional conditions such as:

- wet tissue;
- blood-like fluid;
- different working distances;
- strong and low illumination;
- static and moving tools;
- camera motion;
- partial and severe occlusion.

ToF/LiDAR streams may be used as dynamic visible-surface references only where appropriate, but they will not be treated as unbiased ground truth. If a ToF/LiDAR sensor is part of the comparison, its own measurements will not be used as the absolute reference for that comparison.

4.3 v2: Manipulation and Contact-Aware Extension

The v2 benchmark is a longer-term extension. It may include:

- tool trajectories;
- robot state;
- force or contact sensing;
- tool-tip localization;
- tool-tissue distance labels;
- explicit deformation tracking;
- suturing or manipulation primitives.

This stage is not required for the initial MICCAI 2027 target. It becomes relevant only after the v0/v1 measurement pipeline is stable.

5 Phase 1 Scope

Phase 1 is designed as a controlled feasibility and de-risking experiment. It should remain deliberately narrow. The goal is to test whether the sensing, calibration, reference-geometry, and annotation pipeline is feasible before scaling toward a full benchmark paper.

5.1 In Scope

Phase 1 will focus on:

- ex-vivo porcine tissue as an open-wound surrogate;
- two or three primary sensing modalities;
- fiducial-based multi-camera calibration;
- static reference geometry for selected wound states;
- visible-surface reconstruction;
- core semantic labels: wound region, wound edge, tool, hand/glove, gauze, blood/fluid, background, and occlusion;
- sensor failure analysis under wetness, blood-like fluid, illumination changes, tools, and occlusion;
- specimen-level repeatability across multiple tissue samples.

5.2 Out of Scope for Phase 1

Phase 1 will not attempt to solve:

- full anatomical tissue taxonomy;
- dense dynamic ground truth for every frame;
- complete dynamic 3D ground truth for hidden tissue deformation;
- robust tool-tip 3D tracking in all frames;

- complete tool–tissue distance benchmarking;
- force or contact-aware manipulation;
- clinical operating-room deployment;
- robot control;
- autonomous suturing;
- clinical claims.

6 Phase 1 Questions

Phase 1 must answer the following practical questions:

1. Can the selected cameras be co-calibrated into a common metric coordinate frame with acceptable drift?
2. Can static reference geometry be obtained reliably for open-wound tissue anchor states?
3. Do passive stereo, active stereo, and ToF/LiDAR sensors fail differently under surgical-like nuisance factors?
4. Are sensor differences larger than calibration and reference uncertainty?
5. Can core labels such as wound, tool, hand, fluid, and occlusion be annotated consistently?
6. Is visible-surface reconstruction a measurable and meaningful target for the next ARGOS stage?
7. Can dynamic clips be evaluated using temporal consistency and anchor-state geometry without claiming dense framewise ground truth?

7 Expected Contributions

The expected contributions of ARGOS-Wound are:

1. **A controlled open-surgery wound benchmark:** a multi-view RGB-D dataset acquired on ex-vivo porcine tissue under controlled surgical-like conditions.
2. **A multi-level ground-truth protocol:** static anchor-state reference scans, fiducial-based calibration, semantic masks, wound-edge annotations, tool labels, and occlusion labels.
3. **A systematic sensor evaluation:** comparison of passive stereo, active stereo, ToF/LiDAR, and multi-view RGB reconstruction under open-surgery-relevant nuisance factors.
4. **A dynamic evaluation protocol:** temporal consistency, visible-surface stability, and failure-mode analysis for deforming clips, without claiming dense dynamic ground truth for every frame.
5. **A failure-aware fusion method:** ARGOS-Fuse, a semantic and uncertainty-aware multimodal fusion framework for robust visible-surface reconstruction and unsafe-region detection.
6. **A release-ready dataset package:** synchronized recordings, calibration files, reference scans, annotations, metadata, splits, baseline code, and evaluation scripts.

8 Methodological Direction: Failure-Aware Open-Wound Perception

The intended methodological contribution of ARGOS-Wound is not only multimodal depth fusion, but **failure-aware wound perception**. In surgical guidance, a perception system should not merely output a reconstructed surface; it should also identify regions where the reconstruction is unreliable, incomplete, occluded, or unsafe to use for downstream robotic decisions.

This motivates the development of **ARGOS-Fuse**, a semantic and uncertainty-aware multimodal fusion framework. The central hypothesis is:

Semantic and uncertainty-aware multimodal fusion improves visible wound-surface reconstruction and failure detection under surgical nuisance factors compared with single-modality and naive-fusion baselines.

Each sensing modality is expected to fail differently. Passive stereo may degrade under low texture, specular highlights, blood, illumination changes, and occlusion. Active stereo may fail on reflective or geometrically ambiguous surfaces. ToF/LiDAR may be more robust to illumination, but can exhibit tissue-dependent bias and artifacts on blood or fluid. A clinically useful system should therefore estimate not only the fused surface, but also a spatial map of reliability.

ARGOS-Fuse will aim to produce three outputs:

1. a fused visible wound-surface reconstruction;
2. a geometry uncertainty map;
3. a validity or unsafe-region mask indicating where perception should not be trusted.

This framing makes the benchmark directly relevant to robotic guidance without overclaiming autonomous surgery. The immediate goal is not to control a robot, but to build a perception layer that can support safe downstream modules such as wound-edge localization, tool–tissue distance estimation, deformation compensation, and action gating.

9 Positioning Relative to Prior Work

Prior work has investigated open-surgery video understanding, hand/tool detection, hand–tool segmentation, surgical workflow recognition, and multi-view recording. Therefore, ARGOS-Wound should not be framed as the first dataset for open surgery. Similarly, prior work has studied dense 3D reconstruction of ex-vivo porcine tissue using compact LiDAR or time-of-flight sensors in endoscopic or minimally invasive settings.

The specific gap targeted by ARGOS-Wound is narrower and more precise. In the reviewed literature, we did not identify a release-ready open-surgery benchmark that jointly provides wound-centric semantic annotations, calibrated multi-view RGB-D sensing, reference surface geometry, hand/tool/occlusion labels, and condition-specific evaluation of 3D reconstruction failure modes.

The intended contribution is therefore not to claim that LiDAR or any single sensor is universally superior, nor that open surgery lacks visual datasets. Instead, the benchmark is designed to quantify when each sensing modality succeeds or fails in open surgical wound scenes and to evaluate whether semantic masking and uncertainty-aware multimodal fusion can improve robustness.

The novelty is framed around:

- open-surgery wound-field perception rather than generic open-surgery video understanding;
- wound-field geometry rather than generic organ surface reconstruction;
- anchor-state geometry for controlled deformation clips;

- suturing-relevant perception tasks;
- severe hand/tool occlusion;
- tissue deformation and wound-edge tracking;
- calibrated multi-view RGB-D sensing;
- reference-geometry-based evaluation;
- visible-surface reconstruction under wetness, blood-like fluid, specularity, and illumination changes;
- a release-oriented dataset and evaluation protocol;
- semantic and uncertainty-aware fusion for visible-surface reconstruction;
- failure-aware perception and unsafe-region detection for future robotic guidance.

10 Experimental Setup

10.1 Tissue Model

The first benchmark will use ex-vivo porcine tissue, preferably pork belly or abdominal wall samples. These samples are suitable because they provide visible layers that approximate several open-surgery perception challenges:

- skin;
- fat;
- fascia-like connective tissue;
- muscle;
- wet and deformable surfaces.

The tissue will be prepared to simulate an open surgical wound suitable for suturing-like manipulation. The setup should allow repeated acquisition under controlled variations of lighting, wetness, blood-like fluid, tool presence, and deformation.

10.2 Sensors

The proposed sensor setup is staged.

10.2.1 Mandatory v0 Sensors

The mandatory v0 setup includes:

- **ZED2 stereo camera:** main passive stereo camera for local RGB and depth reconstruction.
- **GoPro Hero 11 or equivalent global RGB camera:** global workflow and occlusion context camera.
- **Structured-light scanner or high-accuracy 3D scanner:** static geometric reference for selected anchor states, if available.
- **Calibration rig:** ChArUco, AprilTags, or 3D spherical fiducials for defining a common world frame.

10.2.2 v1 Candidate Sensors

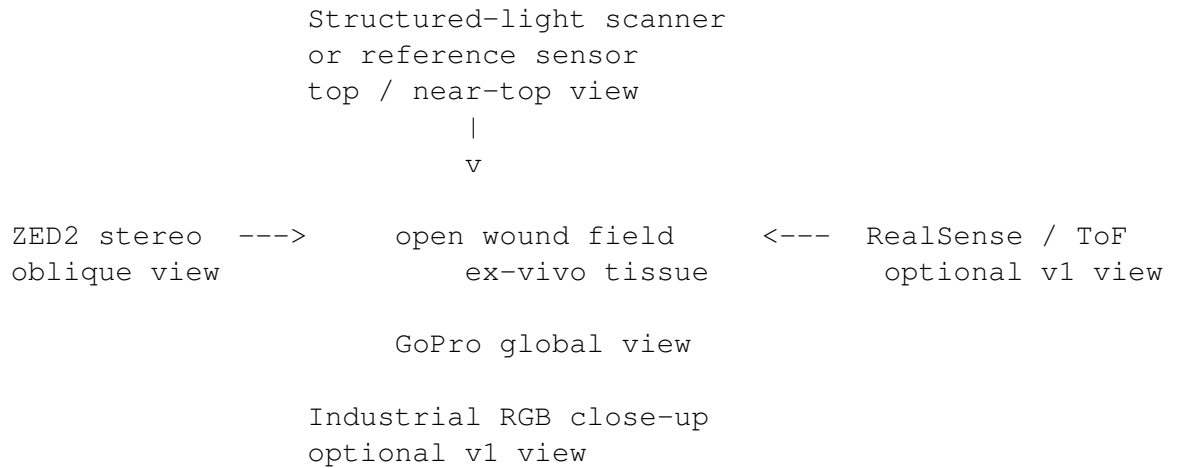
The v1 extension may include:

- **RealSense L515 or Lucid ToF camera:** ToF/LiDAR depth modality to be evaluated, not treated as absolute ground truth.
- **RealSense D455f or D435i:** active stereo RGB-D baseline.
- **Industrial RGB camera:** high-quality close-up RGB stream for segmentation and annotation.

This staged sensor strategy avoids making the first pilot dependent on a complex multi-modal rig before the calibration and anchor-geometry pipeline is validated.

10.3 Camera Geometry

The preferred camera geometry is:



The top or near-top reference acquisition should provide a stable view of the wound surface at anchor states. The oblique sensors should provide clinically realistic viewpoints and reduce single-view occlusion. The GoPro should capture global context, including hands, instruments, camera occlusions, and workflow.

11 Fiducial-Based Calibration Strategy

A central challenge is registering all sensor streams into a common metric coordinate frame. To reduce calibration complexity, ARGOS-Wound will define a local wound coordinate frame using rigid fiducials placed around the tissue sample.

Two types of fiducials will be used:

1. **Rigid calibration fiducials:** AprilTags, ChArUco targets, or 3D spherical markers placed on a rigid frame around the wound. These define the world coordinate frame.
2. **Tissue/deformation fiducials:** optional colored pins, beads, or artificial landmarks placed near the wound edges or on deformable tissue regions. These are not used as a global reference, but to quantify tissue motion and deformation.

The key rule is:

Rigid fiducials define the coordinate frame; tissue fiducials measure deformation.

For each camera C_i , we estimate the transformation from the camera frame to the world frame:

$$T_{W,C_i} \in SE(3),$$

where W denotes the fiducial-defined world frame. A point cloud P_{C_i} acquired in the camera frame is transformed into the common world frame as:

$$P_W = T_{W,C_i} P_{C_i}.$$

If a structured-light scanner is used, the reference mesh is also registered to the world frame using fiducials visible in the scan or through 3D marker correspondences.

12 Data Acquisition Protocol

The acquisition will be organized into short synchronized blocks. This reduces temporal drift, simplifies annotation, and allows repeated calibration checks.

Each dynamic block will be bounded by static anchor states. These anchor states are essential because they allow high-quality reference geometry to be acquired before and after deformation, while the dynamic portion of the clip is evaluated using temporal consistency, visible-surface stability, and failure-mode analysis.

Each block follows the same structure:

1. Start all recordings.
2. Show the block identifier.
3. Trigger a visible synchronization event, such as an LED flash.
4. Record 5–10 seconds of static pre-deformation anchor state.
5. Acquire static reference geometry when feasible.
6. Perform the planned manipulation or acquisition condition.
7. Record 5–10 seconds of static post-deformation anchor state.
8. Acquire post-deformation static reference geometry when feasible.
9. Trigger a final LED synchronization event.

The initial acquisition conditions include:

- clean static wound;
- wet tissue surface;
- blood-like fluid;
- strong illumination;
- low illumination;
- static surgical tools;
- moving surgical tools;
- partial hand/tool occlusion;

- severe occlusion;
- tissue retraction;
- simulated suturing;
- post-suturing or wound-closure state.

13 Specimen Replication

To avoid conclusions based on a single tissue sample, the dataset should include multiple independent tissue specimens. A minimum target is:

$$N_{\text{specimens}} \geq 6,$$

with a preferred target of:

$$N_{\text{specimens}} = 10\text{--}15.$$

Each specimen should undergo repeated acquisition under controlled variations of lighting, wetness, blood-like fluid, tool presence, occlusion, and deformation.

14 Ground-Truth Strategy

The ground-truth strategy is organized as a hierarchy. This is important because deformable open surgical scenes do not provide a single perfect reference for all time points and all tissue regions.

14.1 Tier 1: Static Anchor-State Geometric Reference

For selected static states, a high-accuracy 3D scanner or structured-light system will acquire a reference mesh:

$$M_{\text{GT}}^{(s)}.$$

Here, s denotes a static anchor state. Static reference states may include:

- intact tissue;
- opened wound without tools;
- opened wound with retractors;
- wet or blood-like tissue surface;
- pre-deformation state;
- post-deformation state;
- post-suturing wound state.

These scans provide the strongest geometric reference for evaluating absolute reconstruction accuracy. Dense metric reconstruction claims will primarily be made at these anchor states.

14.2 Tier 2: Dynamic Visible-Surface Reference

If a ToF/LiDAR or RGB-D sensor is recorded continuously, its stream may be used as a dynamic visible-surface reference:

$$D_{\text{ref}}(t),$$

where t denotes time. This reference is not treated as full ground truth for hidden tissue deformation, since it only observes non-occluded surfaces. Instead, it is used as a partial dynamic reference for visible-surface reconstruction and temporal stability analysis.

Importantly, if a ToF/LiDAR sensor is evaluated as a candidate modality, its own stream will not be used as absolute ground truth for that comparison. In such cases, it may only be used as an auxiliary visible-surface reference or for qualitative temporal analysis.

14.3 Tier 3: Dynamic Consistency and Semantic Annotations

Dynamic clips will be evaluated using temporal consistency, visible-surface stability, content masking, and semantic/task annotations rather than dense framewise ground-truth meshes.

A subset of frames will be manually or semi-automatically annotated with semantic masks. Initial classes include:

- wound region;
- wound edge;
- visible tissue;
- skin;
- fat;
- fascia-like tissue;
- muscle;
- blood or fluid;
- surgical tool;
- needle or suture thread;
- hand or glove;
- gauze;
- background;
- uncertain or occluded region.

To reduce annotation noise, two label levels will be maintained:

- **Level 1:** visually reliable labels such as wound, tool, hand, background, fluid, and gauze.
- **Level 2:** surgeon-verified anatomical labels such as fascia, fat, muscle, and wound-edge subregions.

For suturing-related analysis, additional annotations may be generated:

- left and right wound-edge polylines;

- candidate suture entry and exit points;
- tool-tip location;
- needle location, when visible;
- tool-to-tissue distance in selected controlled frames.

Where possible, fiducial markers attached to tools or known tool geometry will be used to recover approximate 3D tool-tip pose.

15 Ground-Truth Policy for Dynamic Deformation

ARGOS-Wound will not claim dense dynamic ground truth for every frame of deforming tissue. This is a deliberate design choice.

The benchmark will instead combine:

- continuous synchronized sensing streams;
- static anchor-state reference geometry;
- semantic and occlusion masks;
- temporal consistency metrics;
- visible-surface completeness metrics;
- sensor failure-mode analysis.

Thus, dynamic clips will be evaluated in two complementary ways:

1. **Metric evaluation at anchor states:** dense point-to-surface accuracy is computed against high-quality reference geometry before and after deformation.
2. **Temporal evaluation during motion:** reconstruction stability, visible-surface consistency, occlusion robustness, and uncertainty-error correlation are evaluated over time.

This design avoids the unrealistic assumption that a complete true deforming mesh is available at every frame, while still allowing the benchmark to evaluate dynamic perception.

16 Benchmark v1 Annotation Scope

To reduce annotation noise and make the first release realistic, ARGOS-Wound will distinguish between core v1 labels and stretch labels.

16.1 Core v1 Labels

The first release will prioritize labels that are visually reliable, surgically meaningful, and feasible to annotate consistently:

- wound region;
- visible wound edge;
- exposed visible tissue;

- surgical tool;
- hand or glove;
- gauze;
- background;
- blood or fluid;
- occlusion state;
- calibration markers and valid evaluation region.

These labels are sufficient to support the first benchmark tasks: wound localization, wound-edge detection, hand/tool masking, occlusion analysis, and visible-surface reconstruction.

16.2 Stretch Labels

The following labels are valuable but will be treated as stretch goals or expert-verified subsets:

- skin;
- fat;
- fascia-like tissue;
- muscle;
- needle;
- suture thread;
- tool-tip keypoints;
- candidate suture entry and exit points;
- tool-to-tissue distance labels.

These labels require stronger expert validation and may not be consistently visible in all frames. They will therefore be released only where annotation quality is sufficient.

17 3D Evaluation Pipeline

For each sensor and acquisition condition, the reconstructed point cloud is transformed into the common world frame and compared against the reference geometry.

Given a reconstructed point cloud:

$$P = \{p_i\}_{i=1}^N,$$

and a reference mesh:

$$M_{GT},$$

we compute a signed point-to-surface error for each point. For each reconstructed point p_i , the closest local triangle on the reference mesh is identified. Let q_i be the closest point or local triangle center on the reference surface and let n_i be the corresponding surface normal. The signed error is defined as:

$$e_i = (p_i - q_i)^\top n_i.$$

Positive values indicate that the reconstructed point is closer to the camera than the reference surface, while negative values indicate that it is farther away.

18 Evaluation Metrics

For each static or quasi-static anchor condition, n consecutive frames are extracted:

$$\{P_t\}_{t=1}^n.$$

The signed error field is computed for each frame and used to derive three primary metrics.

18.1 Depth Accuracy

Depth accuracy measures the systematic signed depth bias:

$$A_i = \frac{1}{n} \sum_{t=1}^n e_i(t).$$

This metric captures whether a sensor consistently estimates the tissue surface as closer or farther than the reference geometry.

18.2 Temporal Variability

Temporal variability measures frame-to-frame instability:

$$V_i = \sqrt{\frac{1}{n-1} \sum_{t=1}^n (e_i(t) - A_i)^2}.$$

This metric is critical for robotic applications, since a depth estimate that is accurate on average but unstable over time may be unsafe for motion planning or tool–tissue distance regulation.

18.3 Shape Precision

Shape precision measures local shape distortion independently of global depth bias. Let $\bar{A}$ be the spatial mean of the depth accuracy field over the evaluated tissue region. Shape precision is defined as:

$$S_i = |A_i - \bar{A}|.$$

This separates global depth shift from local geometric distortion. A sensor may have a constant depth bias but still preserve wound shape accurately; conversely, it may have low mean bias but poor local geometry.

18.4 Benchmark Endpoints

ARGOS-Wound will separate primary, secondary, dynamic, task-specific, and reliability endpoints.

18.4.1 Primary Geometry Endpoint

The primary geometry endpoint is the masked point-to-surface error on the common visible tissue region at static anchor states. It will be reported using robust summary statistics:

- median absolute error;
- interquartile range;
- 90th and 95th percentile error;
- signed bias.

18.4.2 Secondary Geometry Endpoints

Secondary geometry endpoints include:

- temporal variability of depth estimates;
- visible-surface completeness;
- signed depth bias;
- local shape distortion after removing global depth offset;
- failure rate under occlusion or specular reflection.

18.4.3 Dynamic Clip Endpoints

Dynamic clip endpoints include:

- temporal surface consistency;
- drift between pre- and post-deformation anchor states;
- visible-surface stability;
- occlusion recovery;
- sensor dropout rate;
- uncertainty-error correlation over time.

18.4.4 Task-Specific Endpoints

Task-specific endpoints include:

- wound-edge localization error;
- wound-region segmentation accuracy;
- hand/tool segmentation accuracy;
- occlusion classification accuracy;
- tool-tip localization error, where available;
- tool-to-tissue distance error in controlled frames.

18.4.5 Reliability Endpoints

If uncertainty estimates are produced, reliability will be evaluated using:

- calibration curves;
- expected calibration error;
- uncertainty-error correlation;
- selective prediction curves;
- error detection performance under occlusion.

18.4.6 Unsafe-Region Detection Endpoints

Since the intended downstream use is robotic guidance, ARGOS-Wound will explicitly evaluate whether the system can detect regions where perception is unreliable. Unsafe-region detection will be evaluated using:

- AUROC for high-error region detection;
- AUPRC under class imbalance;
- F1 score at selected operating points;
- false-negative rate for unsafe regions;
- selective reconstruction error after rejecting uncertain regions;
- coverage versus error trade-off.

This endpoint is critical because a surgical perception system should avoid silently producing confident but wrong geometry in regions affected by blood, specularity, occlusion, or sensor dropout.

19 View-Field and Content Masking

Different sensors observe different fields of view. Therefore, evaluation will be performed only on regions that are visible to the relevant sensors and included in the reference geometry.

Let Ω_i denote the valid visible region for sensor i , and let Ω_{common} denote the shared visible region across selected sensors. Metrics will be computed either on Ω_i for sensor-specific evaluation or on Ω_{common} for direct sensor comparison.

Content masks will be used to exclude non-target structures:

$$\Omega_{\text{eval}} = \Omega_{\text{visible}} \cap \Omega_{\text{tissue}} \setminus (\Omega_{\text{tool}} \cup \Omega_{\text{hand}} \cup \Omega_{\text{marker}} \cup \Omega_{\text{background}}).$$

This ensures that geometric evaluation reflects reconstruction of the tissue surface rather than tools, hands, fiducials, or background.

20 Spatial and Semantic Pooling

Dense pixel-level or point-level errors are highly spatially correlated. Treating all pixels or points as independent samples would artificially inflate statistical significance. To avoid pseudoreplication, error fields will be spatially pooled into non-overlapping regions or semantically meaningful regions.

Spatial pooling may use a regular grid:

$$G = \{g_1, g_2, \dots, g_K\},$$

where each cell aggregates the mean signed error, temporal variability, and shape precision over a local surface region.

In addition, semantic pooling will be considered for surgically meaningful regions:

- wound edge;
- wound bottom;
- exposed fat;
- fascia-like tissue;

- muscle;
- tool-near region;
- occluded or partially occluded region.

This allows reconstruction errors to be analyzed not only globally, but also in terms of surgical relevance.

21 Statistical Analysis

A key limitation of prior studies is the risk of pseudoreplication, where spatial tiles or pixels are treated as independent biological samples. ARGOS-Wound will explicitly distinguish between biological replication, acquisition-level repetition, temporal frames, and spatial samples.

The primary independent experimental unit will be the tissue specimen:

SpecimenID.

A mixed-effects model will be used, with fixed effects such as:

- sensor type;
- tissue condition;
- lighting condition;
- wetness or blood-like fluid;
- occlusion level;
- motion state;
- tool presence.

Random effects may include:

- specimen ID;
- acquisition block;
- camera view;
- spatial region or tile.

A possible model form is:

$$y \sim \text{Sensor} + \text{TissueCondition} + \text{Lighting} + \text{Occlusion} + \text{Sensor} : \text{TissueCondition} + (1|\text{SpecimenID}) + (1|\text{BlockID}) + (1|\text{CameraView}) + (1|\text{SpatialRegion})$$

Where appropriate, effect sizes and confidence intervals will be reported in addition to p -values. Where the number of independent specimens is insufficient for strong inferential claims, results will be presented as descriptive benchmark trends rather than broad biological conclusions.

22 Uncertainty Budget

To make millimeter-scale claims interpretable, the evaluation will include an uncertainty budget. Sources of uncertainty include:

- reference scanner accuracy;
- camera intrinsic calibration error;
- camera-to-world registration error;
- fiducial localization error;
- synchronization error;
- point-cloud alignment error;
- annotation variability;
- temporal deformation during acquisition.

The final reconstruction errors will be interpreted relative to this reference uncertainty. In particular, differences smaller than the estimated calibration and reference uncertainty will not be overinterpreted.

23 Calibration and Annotation Quality Assurance

23.1 Calibration QA

Since the benchmark relies on millimeter-scale geometric comparison, calibration quality will be reported as a first-class result. The benchmark will report:

- camera intrinsic reprojection error;
- stereo epipolar error, where applicable;
- camera-to-world fiducial reprojection error;
- 3D marker registration error;
- reference scan registration residual;
- block-level pose drift;
- synchronization uncertainty.

At the beginning and end of each acquisition block, the fiducial frame will be recorded to estimate whether camera poses changed during the block. Blocks with excessive drift will either be corrected or excluded from metric evaluation.

23.2 Annotation QA

Because wound and tissue labels can be ambiguous, ARGOS-Wound will include an explicit annotation quality assurance protocol. A subset of frames will be independently annotated by at least two annotators. Disagreements will be resolved through adjudication, and anatomical labels such as fascia-like tissue, fat, muscle, or clinically meaningful wound edges will be reviewed by a surgical expert where possible.

For mask-based labels, inter-annotator agreement will be reported using Dice coefficient and Intersection-over-Union. For wound-edge polylines, agreement will be reported using boundary distance metrics, such as average symmetric surface distance or tolerance-based boundary F-score. Ambiguous regions caused by blood, specularity, occlusion, or poor visibility will be marked as uncertain rather than forced into a class.

24 Baseline Methods

ARGOS-Wound will include multiple baseline methods to avoid conclusions based on a single reconstruction pipeline.

24.1 Depth and Reconstruction Baselines

The initial baselines include:

- ZED2 SDK depth reconstruction;
- RealSense D435i/D455f RGB-D reconstruction;
- RealSense L515 or Lucid ToF reconstruction;
- RAFT-Stereo or equivalent learned stereo baseline;
- classical stereo baseline, such as semi-global matching;
- simple multi-view RGB reconstruction baseline;
- temporal fusion baseline.

24.2 Segmentation Baselines

Semantic segmentation baselines include:

- Segment Anything or SAM2-style zero-shot prompting;
- SegFormer or Mask2Former fine-tuned on annotated frames;
- color/depth heuristic baseline for wound-region detection;
- tool and hand segmentation baseline.

24.3 Fusion and Failure-Detection Baselines

Fusion baselines include:

- single best sensor selected globally;
- simple average fusion;
- median depth fusion;
- sensor-confidence-only fusion;
- semantic-mask-only fusion;
- uncertainty-only fusion without semantic context;
- ARGOS-Fuse without temporal consistency;
- ARGOS-Fuse without semantic features;
- ARGOS-Fuse without uncertainty calibration.

Failure-detection baselines include:

- thresholding sensor confidence;
- thresholding inter-sensor depth disagreement;
- thresholding temporal inconsistency;
- semantic invalid-region masking;
- learned unsafe-region prediction using semantic and geometric cues.

25 Proposed Method: ARGOS-Fuse

Beyond dataset creation and sensor evaluation, we will develop **ARGOS-Fuse**, a failure-aware multi-modal fusion method for open-wound reconstruction.

The goal of ARGOS-Fuse is not simply to average depth maps from multiple sensors. Instead, the method will estimate which modality should be trusted at each location and time, using geometry, sensor confidence, semantic context, temporal consistency, and local failure cues.

The method will take as input:

$$I_{\text{RGB}}, \quad D_{\text{stereo}}, \quad D_{\text{active}}, \quad D_{\text{ToF}}, \quad C_{\text{sensor}}, \quad M_{\text{semantic}}, \quad M_{\text{occlusion}},$$

where I_{RGB} is the RGB image, D_{stereo} is passive stereo depth, D_{active} is active-stereo RGB-D depth, D_{ToF} is ToF/LiDAR depth, C_{sensor} denotes sensor-provided confidence maps when available, M_{semantic} is a semantic mask, and $M_{\text{occlusion}}$ is an estimated occlusion or invalid-region mask.

The output will be:

$$\hat{S}(t), \quad U_{\text{geom}}(t), \quad M_{\text{unsafe}}(t),$$

where $\hat{S}(t)$ is the reconstructed visible wound surface, $U_{\text{geom}}(t)$ is a geometry uncertainty map, and $M_{\text{unsafe}}(t)$ identifies regions where the reconstruction should not be trusted for downstream robotic guidance.

The fusion method will estimate reliability weights:

$$w_k(x, t),$$

for each sensor modality k at pixel or surface location x and time t . The fused depth may be expressed as:

$$\hat{D}(x, t) = \frac{\sum_k w_k(x, t) D_k(x, t)}{\sum_k w_k(x, t) + \epsilon}.$$

The reliability weights will depend on factors such as:

- sensor confidence;
- semantic class;
- specularly or saturation;
- depth discontinuity;
- temporal consistency;
- occlusion probability;
- inter-sensor disagreement;
- proximity to surgical tools;

- tissue type, when available.

This allows the system to avoid naive fusion and instead estimate when stereo, active stereo, or ToF/LiDAR should be trusted.

Importantly, ARGOS-Fuse will be evaluated not only by reconstruction error, but also by its ability to detect unreliable regions. A method that slightly improves mean surface error but fails to identify unsafe regions would be less useful for surgical guidance than a method that provides calibrated uncertainty and reliable failure detection.

26 Dataset Release Plan

To maximize reproducibility and benchmark value, the dataset release will include:

- synchronized RGB/RGB-D video streams;
- raw or processed point clouds;
- camera intrinsics and extrinsics;
- calibration target recordings;
- reference 3D scans;
- semantic masks;
- wound-edge annotations;
- occlusion labels;
- acquisition metadata;
- specimen metadata;
- train/validation/test splits;
- baseline predictions;
- baseline code;
- evaluation scripts;
- annotation guidelines;
- dataset documentation and ethical notes;
- license and access terms.

The release will explicitly distinguish between raw data, processed data, and benchmark-ready annotations. If full raw data release is restricted, a processed benchmark subset with calibration, annotations, reference geometry, and evaluation scripts will be released.

27 Go / No-Go Criteria

The Phase 1 experiment will be used to decide whether ARGOS-Wound should scale into a larger benchmark and method-development effort targeting MICCAI 2027.

27.1 Go Criteria

The project should continue to Phase 2 if:

- camera-to-world calibration can be repeated with stable error below a predefined tolerance;
- static anchor-state reference geometry can be acquired reliably for selected wound states;
- at least one sensor provides usable visible-surface reconstruction under clean conditions;
- sensor-specific failure modes are measurable under wetness, blood-like fluid, illumination changes, and occlusion;
- core semantic labels can be annotated with acceptable inter-annotator agreement;
- reconstruction errors are larger than the estimated calibration/reference uncertainty, making sensor comparison meaningful;
- dynamic clips can be evaluated through temporal consistency and anchor-state geometry without requiring dense framewise ground truth;
- the acquisition protocol can be repeated across multiple specimens without excessive setup burden.

27.2 Pause or Pivot Criteria

The project should pause or pivot if:

- calibration drift dominates all geometric measurements;
- reference geometry is too unstable or too difficult to acquire;
- all candidate sensors fail under even clean or moderately realistic conditions;
- tissue and wound labels are too ambiguous for reliable annotation;
- sensor differences are smaller than the uncertainty of the reference system;
- annotation cost is disproportionate to the information gained;
- the setup is too complex to repeat across specimens.

28 Target Venue and Publication Path

The primary publication target is **MICCAI 2027**. This target is appropriate because the work is centered on medical image analysis, surgical data science, 3D reconstruction, dataset design, semantic segmentation, uncertainty estimation, and benchmark-based evaluation.

The intended MICCAI 2027 submission should be based on the v0/v1 version of ARGOS-Wound, not on the full long-term v2 autonomy-oriented extension. The most realistic submission is a benchmark-and-method paper combining anchor-state reference geometry, calibrated multi-view acquisition, sensor failure analysis, core semantic labels, and uncertainty-aware visible-surface reconstruction.

The publication path is staged:

1. **Internal de-risking phase:** validate calibration, reference geometry, sensor behavior, and annotation feasibility on ex-vivo tissue.
2. **v0 benchmark construction phase:** acquire stereo/RGB-centered ARGOS-Wound v0, process the data, annotate core labels, define anchor-state evaluation, and implement initial baselines.

3. **v1 extension phase:** add active stereo and ToF/LiDAR modalities as evaluated sensors, introduce nuisance-factor conditions, and strengthen sensor comparison.
4. **Method-development phase:** implement ARGOS-Fuse and compare it against passive stereo, active stereo, ToF/LiDAR, naive fusion, and failure-detection baselines.
5. **MICCAI 2027 submission:** submit a benchmark-and-method paper if Phase 1 produces sufficient evidence and the dataset is release-ready.

Possible secondary or fallback venues include ICRA 2027 if the robotic perception framing becomes dominant, MICCAI Open Data or workshop tracks for dataset-centered dissemination, and NeurIPS Datasets and Benchmarks / Evaluations and Datasets as a longer-term target if the dataset becomes sufficiently large and broadly reusable.

29 Timeline Toward MICCAI 2027

A preliminary timeline is:

- **May–June 2026:** finalize Phase 1 protocol, select sensors, build the fiducial calibration setup, and perform preliminary ex-vivo acquisitions.
- **July–August 2026:** complete Phase 1 de-risking experiments, including calibration QA, static anchor-state reference geometry tests, and core label feasibility checks.
- **September–October 2026:** acquire ARGOS-Wound v0/v1 across multiple specimens and controlled conditions; process data and compute calibration.
- **November–December 2026:** generate core annotations, implement baseline reconstruction, segmentation, fusion, and failure-detection methods.
- **January 2027:** finalize experiments, ablations, statistical analysis, figures, dataset documentation, and release package.
- **February–March 2027:** prepare and submit the MICCAI 2027 paper, assuming Phase 1 and dataset-release criteria are satisfied.

30 Risk Analysis

The main risks are:

- **Calibration complexity:** multi-sensor calibration may become the main bottleneck. This will be mitigated by using a fixed fiducial-defined world frame and short acquisition blocks.
- **Reference geometry limitations:** static scanner or structured-light geometry may only be available at anchor states. This will be handled explicitly by separating anchor-state metric evaluation from dynamic temporal evaluation.
- **Sensor failure on wet/specular tissue:** ToF, active stereo, and passive stereo may fail differently under blood, wetness, and reflections. This is not only a risk but also a central research target.
- **ToF/LiDAR bias:** ToF/LiDAR may exhibit tissue-dependent depth bias and should not be treated as unbiased ground truth.
- **Insufficient biological replication:** conclusions based on too few tissue specimens would be weak. The protocol therefore targets multiple independent specimens.

- **Annotation cost:** dense semantic labels are expensive. The dataset will use a layered annotation strategy with dense labels on selected frames and lighter labels on larger sequences.
- **Failure-detection ambiguity:** defining unsafe or unreliable regions may require careful thresholding and expert review. This will be mitigated by using high-error regions, occlusion masks, sensor dropout, and uncertainty calibration as complementary definitions.
- **Overclaiming:** the paper must avoid claiming general superiority of one modality. The intended claim is benchmark-based, condition-specific, and failure-aware.
- **Dataset release constraints:** if full raw data release is difficult, a processed benchmark subset with calibration, annotations, and evaluation scripts should still be released.

31 Dependencies and Bottlenecks

Dependency	Risk	Mitigation
Reference scanner	No sufficiently accurate scanner is available	Start with anchor-state RGB-D/ToF reference and seek access to structured-light scanner
Calibration hardware	Multi-camera calibration may be unstable	Use rigid fiducial frame, ChArUco board, and 3D markers
Sensor synchronization	Streams may not be hardware synchronized	Use short blocks, LED sync, and software timestamps
Tissue specimens	Specimen availability may limit replication	Start with minimum viable number, then scale
Annotation labor	Dense labels may be too expensive	Use core labels first and annotate selected frames
Tissue ambiguity	Fine tissue classes may be unreliable	Separate core labels from expert-only labels
Dynamic ground truth	Dense framewise GT may be unrealistic	Use static anchor geometry and dynamic consistency metrics
Failure labels	Unsafe-region definition may be ambiguous	Use multiple definitions: high-error regions, occlusion, dropout, and uncertainty calibration
Data release	Raw data may be difficult to publish	Release processed benchmark subset with calibration and evaluation scripts

Table 1: Main dependencies and mitigation strategies for Phase 1.

32 Success Criteria

The pilot will be considered successful if it produces:

1. a stable multi-sensor acquisition setup;
2. calibrated point clouds in a common world frame;
3. at least one reliable static anchor-state geometric reference protocol;
4. meaningful failure analysis for ZED2, RealSense, and ToF/LiDAR sensors;
5. annotated wound, tool, hand, and occlusion labels on selected frames;

6. baseline reconstruction and segmentation results;
7. a dynamic evaluation protocol based on temporal consistency and anchor-state geometry;
8. preliminary evidence that semantic uncertainty-aware fusion improves robustness or correctly identifies unreliable regions.

33 Phase 1 Decision Outcome

At the end of Phase 1, the project should produce one of three outcomes:

1. **Proceed:** calibration, reference geometry, annotation, and sensor comparison are feasible. The project should scale toward ARGOS-Wound v0/v1 and a publishable benchmark.
2. **Proceed with modification:** the general direction is valid, but one or more components must be changed, such as sensor choice, reference strategy, label scope, or acquisition protocol.
3. **Pause:** the perception problem cannot yet be measured reliably with the available hardware and protocol. The project should delay larger autonomy-facing work until the measurement bottleneck is solved.

34 Claim Boundaries

ARGOS-Wound does not claim that:

- open surgery lacks visual datasets;
- ex-vivo porcine tissue fully reproduces clinical open surgery;
- ToF/LiDAR, stereo, or active stereo is universally superior;
- ToF/LiDAR provides unbiased ground truth;
- dynamic depth streams provide complete ground truth for hidden tissue deformation;
- dense dynamic ground truth is available for every frame;
- uncertainty-aware fusion guarantees safe robotic action;
- failure detection replaces surgeon supervision or clinical validation;
- the first benchmark release solves autonomous suturing.

Instead, ARGOS-Wound claims that:

- wound-centric open-surgery semantic-geometric benchmarking remains underdeveloped relative to MIS/endoscopic benchmarks;
- controlled ex-vivo tissue is a useful first metrology step;
- anchor-state reference geometry can support rigorous evaluation without requiring unrealistic dense dynamic ground truth;
- multi-sensor RGB-D acquisition can reveal modality-specific failure modes;
- semantic masking and uncertainty estimation are necessary for reliable visible-surface reconstruction;

- failure-aware perception is a necessary intermediate step between wound reconstruction and robotic guidance;
- uncertainty and unsafe-region prediction provide a safer evaluation target than reconstruction accuracy alone;
- a calibrated benchmark can support future robotic guidance research.

35 Conclusion

ARGOS-Wound is designed as the first perception-focused benchmark within the broader ARGOS project. Its purpose is to move from a broad vision of autonomous robotic guidance in open surgery to a concrete, measurable, and publishable research problem: local semantic-geometric understanding of the open wound field.

In its internal-proposal form, the goal of ARGOS-Wound is to de-risk the measurement problem. If Phase 1 demonstrates that the scene can be calibrated, reconstructed, annotated, and evaluated reliably, the project can scale toward a release-ready benchmark and uncertainty-aware reconstruction method targeting MICCAI 2027.

The updated strategy is deliberately staged. The first benchmark version will focus on synchronized sensing, fiducial-based calibration, core labels, and high-quality static anchor geometry. Dynamic clips will be evaluated through temporal consistency, visible-surface stability, occlusion recovery, and failure-mode analysis rather than through unrealistic dense framework ground truth.

The central methodological direction is ARGOS-Fuse: semantic and uncertainty-aware multimodal fusion for failure-aware wound perception. The strongest scientific contribution is not only estimating the visible wound surface, but also identifying where that estimate is unreliable and should not be trusted by downstream robotic systems.

The long-term role of ARGOS-Wound is to support downstream modules for robotic suturing, tool-tissue distance estimation, deformation compensation, safety-aware action gating, and autonomous surgical guidance.

A Appendix A: Phase 1 Practical Acquisition Setup Based on Available AAU Equipment

The Phase 1 acquisition will use equipment currently available within the AAU laboratory inventory. The objective is not to deploy the final ARGOS sensing platform, but to build a minimal, stable, and repeatable acquisition setup for ex-vivo porcine wound perception.

The main design principle is:

Use the smallest sensor configuration that can support synchronized acquisition, stereo or RGB-D reconstruction, fiducial-based calibration, and static anchor-state evaluation.

This avoids overloading the first pilot with too many devices, drivers, synchronization requirements, and calibration dependencies.

A.1 Recommended Phase 1 Sensor Configuration

The recommended Phase 1 setup consists of the following devices:

The ZED2 stereo camera should be used as the primary sensor because it provides a complete and practical stereo pipeline, including left/right RGB images, depth maps, and point-cloud export through the Stereolabs SDK. Although it may not provide the strongest hardware-triggered synchronization setup, it is the most pragmatic choice for an initial proof of concept.

Role	Preferred Device	Purpose
Main stereo sensor	ZED2 stereo camera	Primary stereo RGB/depth stream and main reconstruction baseline
Secondary RGB-D sensor	RealSense D435i, D435, or D455f	Independent active-stereo/depth baseline and modality comparison
Side RGB camera	AV Mako G-234C, iDS UI-3140CP-C-HQ, or Point Grey Chameleon3	Lateral context view, segmentation support, and occlusion analysis
Global context camera	GoPro Hero 7 Black, if needed	Wide-angle documentation of the acquisition scene and operator actions
Lighting	Godox LEDP260c, IKEA FLOALT, or SVL white light	Stable and diffuse illumination of the wound field
Calibration target	ChArUco board or checkerboard	Intrinsic/extrinsic calibration and camera pose estimation
Synchronization target	Visible LED flash	Offline temporal alignment across sensor streams
Fiducial frame	ArUco/AprilTag board or physical colored fiducials	Definition of a local wound/world coordinate frame

Table 2: Recommended practical sensor configuration for the first ARGOS-Wound ex-vivo acquisition.

A RealSense D435i, D435, or D455f should be used as a secondary RGB-D sensor. This sensor should not be treated as ground truth. Instead, it should be evaluated as an additional depth modality and compared against the ZED2 under different tissue and illumination conditions.

One industrial RGB camera, such as an AV Mako G-234C, iDS UI-3140CP-C-HQ, or Point Grey Chameleon3, should be used only if it can be tested before the acquisition day. Its role is to provide a lateral high-quality RGB stream for semantic segmentation, hand/tool occlusion analysis, and qualitative validation.

The GoPro Hero 7 Black may be used as a global context camera, but it should not be part of the primary metric reconstruction pipeline because of lens distortion, limited synchronization, and limited scientific control.

A.2 Sensors Not Recommended for the First Pilot

Several available devices are interesting but should not be included in the first ex-vivo acquisition unless there is sufficient preparation time.

- **Bumblebee XB3 Stereo Vision System:** scientifically interesting, but driver compatibility and acquisition stability must be verified before use.
- **Lytro Illum B5:** potentially interesting for light-field imaging, but it would introduce a separate acquisition and processing pipeline.
- **Kinect V1/V2, Pico Flexx, and older ToF devices:** useful for exploratory tests, but likely less reliable for wet, specular, close-range tissue scenes.
- **Hokuyo, rpLidar, and Scanse Sweep:** primarily planar LiDAR systems and not suitable as main wound-surface reconstruction sensors.
- **Thermal cameras:** not relevant for the first geometry-focused benchmark version.
- **360-degree cameras:** useful for documentation but not for calibrated wound-field reconstruction.

The first acquisition should prioritize a stable and repeatable pipeline over broad sensor coverage.

A.3 Minimal Phase 1 Setup

The minimal recommended setup is:

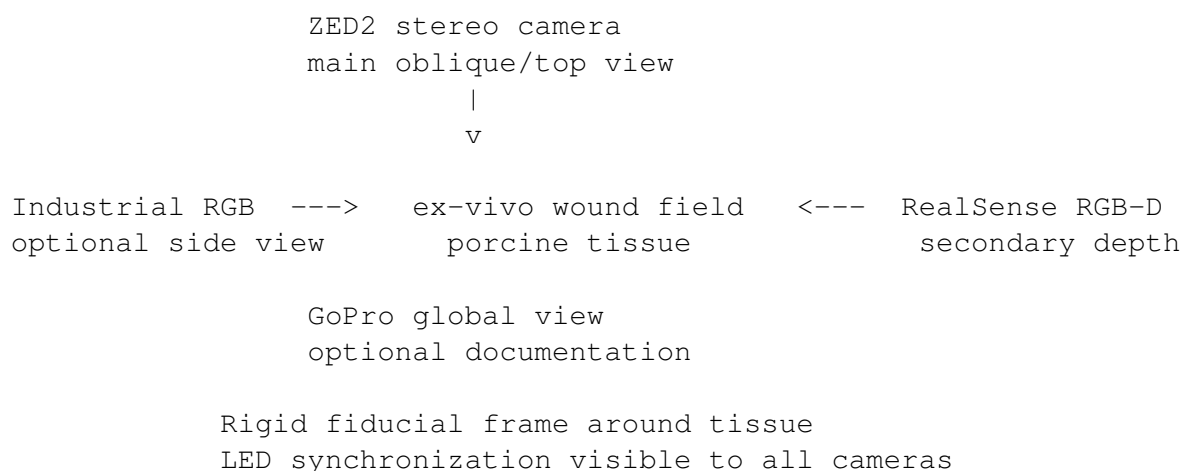
- ZED2 stereo camera above or oblique to the wound field;
- RealSense RGB-D sensor from a second viewpoint;
- visible LED synchronization event;
- ChArUco or checkerboard calibration target;
- rigid fiducial frame around the tissue specimen;
- diffuse lighting;
- synchronized software timestamp logging;
- external SSD storage for all acquired streams.

The first pilot should be considered successful if this minimal setup can produce:

- stable stereo/RGB-D recordings;
- frame-level timestamps;
- visible LED synchronization events;
- calibrated camera intrinsics;
- estimated camera-to-world transformations;
- reconstructed point clouds in a common coordinate frame;
- static anchor-state point clouds suitable for preliminary comparison.

A.4 Physical Camera Layout

The preferred physical layout is:



The ZED2 should have the clearest and most stable view of the wound. The RealSense should observe the same region from a slightly different viewpoint. The industrial RGB camera, if used, should provide a lateral view with minimal occlusion. The LED synchronization marker must be visible in all image streams.

A.5 Rigid Fiducial Frame

A rigid fiducial frame should be placed around the porcine tissue specimen. The frame should not move during acquisition. It should contain at least four visible markers, with six to ten markers preferred.

Possible fiducials include:

- printed ArUco or AprilTag markers;
- ChArUco board corners;
- colored spherical markers;
- rigid beads or small 3D markers;
- hybrid 2D–3D fiducial targets.

The role of this frame is to define the local wound/world coordinate system. Tissue deformation markers, if used, should be treated separately. The key rule is:

Rigid fiducials define the world frame; tissue fiducials measure tissue motion.

A.6 Synchronization Strategy for Phase 1

Hardware-level synchronization is not required for the first pilot. Instead, the Phase 1 acquisition will use a hybrid offline synchronization strategy based on:

1. software or SDK timestamps for every acquired frame;
2. short acquisition blocks to reduce clock drift;
3. visible LED flashes at the beginning and end of each block.

Each recording block should follow this pattern:

1. start all sensors;
2. wait until all streams are stable;
3. trigger three visible LED flashes;
4. record the target acquisition condition;
5. trigger three visible LED flashes again;
6. stop all sensors;
7. immediately check that files and timestamps were saved correctly.

Each sensor stream must save a timestamp file:

```
frame_id,timestamp_host,timestamp_device,filename
000001,0.000000,123456.000000,left_000001.png
000002,0.033421,123456.033421,left_000002.png
000003,0.066825,123456.066825,left_000003.png
```

After acquisition, an offline synchronization table will be generated:

```
sync_id,t_global,zed_left,zed_right,realsense_rgb,realsense_depth,side_rgb
000001,0.000,000120,000120,000087,000087,000044
000002,0.033,000121,000121,000088,000088,000045
000003,0.066,000122,000122,000089,000089,000046
```

This table will define which frames belong to the same approximate physical time point.

A.7 Phase 1 Equipment Checklist

Before the ex-vivo acquisition, the following equipment should be prepared.

A.7.1 Sensors and Computers

- ZED2 stereo camera and cable;
- RealSense D435i, D435, or D455f and cable;
- optional industrial RGB camera and lens;
- optional GoPro Hero 7 Black for global context;
- acquisition laptop with all SDKs installed;
- laptop power supply;
- external SSD with sufficient free space;
- powered USB hub;
- USB-C/USB-A adapters;
- long and reliable data cables;
- power strip and extension cable.

A.7.2 Mounting and Stability

- tripods;
- clamps;
- magic arms or articulated camera arms;
- camera mounting adapters;
- tape;
- cable ties;
- weights or stabilizers for tripods;
- labels for identifying sensors and cables.

A.7.3 Calibration and Fiducials

- rigid ChArUco board;
- backup checkerboard;
- printed ArUco or AprilTag markers;
- colored spherical fiducials or beads;
- rigid frame or tray around the tissue specimen;
- ruler or caliper;
- marker pen and labels.

A.7.4 Synchronization

- visible LED or flashlight;
- optional Arduino or Raspberry Pi Pico for controlled LED flashing;
- power bank or battery;
- backup manual flashlight.

A.7.5 Lighting

- Godox LEDP260c or equivalent LED panel;
- IKEA FLOALT or SVL white light as backup;
- light diffuser or umbrella, if available;
- tape or clamps for lighting placement.

A.7.6 Ex-Vivo Handling

- gloves;
- absorbent paper;
- protective table covering;
- stable tray for the tissue;
- tools for manipulation;
- saline or water spray, if needed;
- waste container according to laboratory rules.

A.8 Pre-Acquisition Test Requirements

The ex-vivo acquisition should not be performed until the following tests have been completed.

1. The ZED2 records left images, right images, depth maps, and timestamps.
2. The RealSense records RGB, depth, and timestamps.
3. The LED synchronization flash is visible in all active video streams.
4. The ChArUco or checkerboard target is detected in the camera images.
5. The acquisition laptop can record all selected streams without dropping frames.
6. The external SSD can sustain the required write speed.
7. The folder structure and metadata files are prepared before the acquisition.
8. At least one short test acquisition has been processed offline.

If these requirements are not met, the pilot should be simplified rather than expanded.

Block ID	Condition
acq_001	Clean static wound, no tools, no hand occlusion
acq_002	Wet tissue surface
acq_003	Blood-like fluid or colored liquid
acq_004	Static surgical tools in the field
acq_005	Moving tool occlusion
acq_006	Hand/glove occlusion
acq_007	Tissue retraction or deformation
acq_008	Simulated suturing motion
acq_009	Post-suturing or partially closed wound state

Table 3: Suggested acquisition blocks for the first ex-vivo porcine tissue session.

A.9 Acquisition Blocks

The first ex-vivo session should be divided into short acquisition blocks:

Each block should be short, documented, and independently synchronized. This makes the dataset easier to debug, process, and annotate.

A.10 Data Folder Structure

The recommended folder structure for the first pilot is:

```

argos_wound_pig_trial_001/
  README.md
  metadata.yaml
  acquisition_log.md

  calibration/
    charuco_images/
    zed_calibration/
    realsense_calibration/
    extrinsics/
    fiducial_frame/

  sync/
    led_events.csv
    sync_table.csv

  setup_photos/

  acq_001_static_clean/
    zed/
      left/
      right/
      depth/
      point_cloud/
      timestamps.csv
    realsense/
      rgb/
      depth/
      point_cloud/
      timestamps.csv

```



```

    side_rgb/
        frames/
        timestamps.csv
    notes.md

acq_002_wet_tissue/
    ...

acq_003_blood_like_fluid/
    ...

acq_004_static_tools/
    ...

acq_005_moving_tools/
    ...

acq_006_hand_occlusion/
    ...

acq_007_tissue_deformation/
    ...

acq_008_simulated_suturing/
    ...

reference_geometry/
    anchor_state_scans/
    processed_reference_meshes/

evaluation/
    reconstruction_errors/
    figures/
    qa_reports/

```

This folder structure should be created before the acquisition day. Data should not be stored in an ad hoc way.

A.11 Reference Geometry Limitation

The current AAU equipment inventory contains several RGB-D and depth sensors, but it is not yet clear whether a high-accuracy structured-light or metrology-grade 3D scanner is available. Therefore, the first pilot should distinguish between:

1. **true reference geometry**, if a high-quality scanner is available;
2. **pseudo-reference geometry**, obtained by averaging static RGB-D/depth measurements across many frames;
3. **cross-sensor comparison**, where ZED2 and RealSense reconstructions are compared qualitatively and quantitatively without claiming either as absolute ground truth.

The first pilot should not claim millimeter-scale ground truth unless the reference scanner and calibration uncertainty support such a claim.

A.12 Phase 1 Practical Success Criterion

The first ex-vivo acquisition should be considered successful if it produces:

1. synchronized ZED2 and RealSense streams;
2. a visible LED synchronization event in all streams;
3. valid timestamp files for every sensor;
4. stable camera placement throughout each block;
5. fiducials visible around the wound field;
6. at least one static clean wound sequence;
7. at least one wet or blood-like condition sequence;
8. at least one occlusion or tool-interaction sequence;
9. preliminary point clouds reconstructed from the main sensors;
10. enough metadata to reproduce the setup.

If these outputs are obtained, Phase 1 will have achieved its main goal: validating that open-wound perception data can be acquired in a controlled, calibrated, and processable way.